Deep-Research Audit: Nonlinear Dynamics of a Sine-Modulated Reaction-Diffusion-Renewal System

1. Executive Introduction and Methodological Demarcation

This report provides a rigorous, exhaustive audit and peer-level critique of the reaction-diffusion-renewal system defined by the governing non-linear partial differential equation $\frac{\partial U}{\partial t} = D\nabla^2 U + \lambda U^2 \sin(\alpha U) + \Gamma(U)$. The analysis is conducted strictly under the locked parameter regime specified as $D=0.12$, $\lambda=0.45$, $\alpha=\pi$, and subject to an implementation-level non-negativity numerical clamp ($U \geq 0$). The structural complexity of this equation arises from the interplay between a non-linear polynomial growth term (U^2) and a periodic trigonometric modulation ($\sin(\pi U)$). Together, these terms sculpt a highly intricate, multiscale landscape characterized by profound bistability, metastable trapping, and critical multidimensional nucleation thresholds.

To conduct a mathematically sound critique of this system, it is absolutely mandatory to enforce strict methodological boundaries. A pervasive analytical flaw in contemporary non-linear dynamics literature is the erroneous, often implicit, conflation of localized continuous models with their simulated discrete counterparts. Therefore, this audit explicitly distinguishes throughout whether its mathematical conclusions apply to:

1. **The local ordinary differential equation (ODE) kinetics:** The zero-dimensional, purely temporal dynamics representing the isolated reaction terms without spatial coupling.
2. **The continuous partial differential equation (PDE) interpretation:** The spatio-temporal evolution in continuous infinite domains, governing analytical front propagation, true topological heteroclinic orbits, and exact mean curvature flows.
3. **The discretized finite-grid implemented system:** The actual numerical manifestation of the system where continuous translational invariance is fundamentally broken, introducing artifactual friction, wave pinning, lattice-induced anisotropy, and artificial boundary behaviors.

Furthermore, a central requirement of this audit is to classify the nature of the structures observed at $U \approx 1$, $U \approx 2$, and $U \approx 3$. It must be explicitly stated at the outset: **These structures are exact mathematical fixed-points of the unperturbed local ODE kinetics.** They are derived directly and analytically from the roots of the trigonometric operator $\sin(\pi U) = 0$. Consequently, they operate as an effective phase-space interpretation of the continuous PDE, defining the bounds of the heteroclinic traveling waves. Finally, they manifest as implementation-level empirical observations in the discretized finite-grid system, where their exact positions may be subject to minor lattice-induced shifting or numerical blurring depending on the spatial resolution.

The following sections systematically dismantle the system's phase space, stability criteria, gradient flow properties, classical nucleation constraints, and susceptibility to propagation failure. Hype regarding "sustained chaos" or "stable chaos" is explicitly excluded; mathematical evidence demonstrates that such gradient-like spatial formulations, in the absence of delayed or highly specific cross-coupled renewal terms, inexorably collapse to steady-state attractors characterized by strictly negative maximum Lyapunov exponents. Any chaotic persistence in

such a system is demonstrably false, generally arising from numerical instability rather than intrinsic continuum topology.

2. Local ODE Kinetics: Topological Phase Space and Fixed-Point Rigor

Before introducing the spatial diffusion operator ($D\nabla^2 U$) or the structurally unspecified renewal operator ($\Gamma(U)$), the foundational topology of the system must be rigorously evaluated through the lens of its local ODE kinetics. By stripping away spatial dimensions, we isolate the reaction kinetics: $\frac{dU}{dt} = f(U) = 0.45 U^2 \sin(\pi U)$.
This isolation allows for an exact analytical treatment of the system's equilibria. The roots of the kinetic function occur wherever $\sin(\pi U) = 0$, which dictates that every integer value of U is an equilibrium point. However, the presence of the U^2 term fundamentally alters the local phase space near the origin. To determine the linear stability of these exact mathematical fixed points in the local ODE kinetics, we must evaluate the Jacobian derivative $J(U) = f'(U) = 0.45 [2U \sin(\pi U) + \pi U^2 \cos(\pi U)]$.
The evaluation of the Jacobian at the critical integers yields the following topological classifications:

- **At the origin (U=0):** $J(0) = 0.45 [0 + 0] = 0$. Because the leading eigenvalue is exactly zero, U=0 is a non-hyperbolic fixed point. Linear stability analysis fails here. The dynamics in the immediate neighborhood $U \to 0^+$ are governed by the leading-order Taylor series expansion $f(U) \approx 0.45 \pi U^3$. Because the exponent is greater than one, a trajectory starting at a very small positive value $U_0 \ll 1$ will diverge from zero, but it will do so algebraically rather than exponentially. This non-hyperbolic repulsion dictates that the basin of attraction toward U=1 from the left boundary is highly lethargic and exceedingly vulnerable to numerical noise, a factor that will heavily penalize the discretized finite-grid implementation.
- **At the first structural node (U=1):** $J(1) = 0.45 [0 + \pi(1)^2 \cos(\pi)] = -0.45\pi \approx -1.4137$. Because J(1) is strictly negative, U=1 is a stable mathematical fixed point of the local ODE. Small perturbations around U=1 will decay exponentially, establishing this node as the center of the first attractive basin.
- **At the second structural node (U=2):** $J(2) = 0.45 [0 + \pi(2)^2 \cos(2\pi)] = 1.8\pi \approx 5.6548$. Because J(2) is strictly positive and quite large, U=2 is an intensely unstable mathematical fixed point of the local ODE. In the zero-dimensional topology, it acts purely as a local separatrix or watershed threshold, strictly dividing the basins of attraction for U=1 and U=3.
- **At the third structural node (U=3):** $J(3) = 0.45 [0 + \pi(3)^2 \cos(3\pi)] = -4.05\pi \approx -12.7234$. Because J(3) is strictly negative and possesses an absolute magnitude significantly larger than J(1), U=3 is a massively stable mathematical fixed point of the local ODE, exhibiting a highly aggressive restorative force against local perturbations.

ODE Fixed Point (U^*)	Analytical Jacobian $J(U^*)$	Topological Classification	Asymptotic Behavior
U = 0	0	Non-hyperbolic Repellor	Algebraic divergence ($\sim U^3$)
U = 1	$-0.45\pi \approx -1.41$	Stable Node	Exponential decay
U = 2	$+1.8\pi \approx +5.65$	Unstable Node (Separatrix)	Exponential divergence
U = 3	$-4.05\pi \approx -12.72$	Stable Node	Rapid exponential decay

The topological hierarchy established by the local ODE kinetics proves that the $U \approx 1$, $U \approx 2$, and $U \approx 3$ structures are not arbitrary artifacts. They are rigid mathematical constraints inherent to the underlying trigonometric kinetics. However, while the ODE identifies both U=1 and U=3 as locally stable nodes, it is fundamentally blind to the broader thermodynamic hierarchy between them. To understand which state dominates when spatial coupling is introduced, we must transition to an energy-based continuous PDE analysis.

3. The Exact Effective Potential: Gradient Flow and Thermodynamic Hierarchy

The use of an "effective potential" is frequently abused as a loose heuristic in applied mathematics and complex systems modeling. However, the explicit prompt constraint demands that this concept be used only if mathematically justified. In this specific unperturbed formulation—where $\Gamma(U)$ is temporarily set to zero and no cross-diffusion or advective vectors are present—the concept is strictly and mathematically rigorous for both the local ODE kinetics and the continuous PDE interpretation.

The mathematical justification arises because the reaction term $f(U)$ depends solely on the scalar state variable U. Consequently, the continuous PDE $\frac{\partial U}{\partial t} = D\nabla^2 U + f(U)$ can be rewritten identically as an L^2-gradient descent flow: $\frac{\partial U}{\partial t} = -\frac{\delta \mathcal{E}}{\delta U}$, where the Lyapunov free-energy functional is defined as $\mathcal{E}[U] = \int \left(\frac{D}{2}|\nabla U|^2 + V(U) \right) dx$.

The exact effective potential density $V(U)$ is mathematically defined by the negative antiderivative of the kinetic function: $V(U) = -\int f(U)\, dU$. Because the system is a gradient flow, the free-energy functional must monotonically decrease over time ($\frac{d\mathcal{E}}{dt} \leq 0$). This absolute constraint mathematically ensures that the continuous PDE interpretation will inexorably seek local or global minima in $V(U)$, categorically eliminating the possibility of sustained chaos, strange attractors, or limit cycles.

We derive the exact analytic potential by integrating the locked parameters:

Utilizing integration by parts twice, we obtain the exact closed-form expression for the effective potential:

Evaluating this exact effective potential at the mathematical fixed points of the local ODE reveals the true thermodynamic hierarchy of the continuous PDE interpretation:

- **Setting the baseline (U=0):** $V(0) = -0.45 \left(\frac{2}{\pi^3} \right) \approx -0.0290$.
- **At the first basin (U=1):** $V(1) = -0.45 \left[-\left(\frac{1}{\pi} - \frac{2}{\pi^3} \right) (-1) \right] = 0.45 \left(\frac{2}{\pi^3} - \frac{1}{\pi} \right) \approx -0.1142$.
- **At the barrier (U=2):** $V(2) = -0.45 \left[-\left(\frac{4}{\pi} - \frac{2}{\pi^3} \right) (1) \right] = 0.45 \left(\frac{4}{\pi} - \frac{2}{\pi^3} \right) \approx +0.5439$.
- **At the second basin (U=3):** $V(3) = -0.45 \left[-\left(\frac{9}{\pi} - \frac{2}{\pi^3} \right) (-1) \right] = 0.45 \left(\frac{2}{\pi^3} - \frac{9}{\pi} \right) \approx -1.2601$.

Phase State (U)	Exact Effective Potential V(U)	Thermodynamic Classification	Transition Energy (ΔV)
U = 1	≈ -0.1142	Metastable False Vacuum	$\Delta V_{1 \to 2} \approx 0.6581$
U = 2	$\approx +0.5439$	Nucleation Barrier Peak	N/A
U = 3	≈ -1.2601	Stable True Vacuum	$\Delta V_{3 \to 2} \approx 1.8040$

This rigorous mathematical derivation exposes a profound topological asymmetry that the local ODE kinetics alone could not reveal. While both U=1 and U=3 are stable attractors in the zero-dimensional ODE, the effective potential of the continuous PDE interpretation proves that

$V(3) \ll V(1)$. In the rigorous language of phase transitions, U=3 represents the **true vacuum** (global minimum) of the system, whereas U=1 is merely a metastable **false vacuum**.
The unstable fixed point at U=2 forms a massively asymmetric energy barrier. The potential energy required to perturb the system from the false vacuum (U=1) over the barrier to trigger an avalanche is $\Delta V_{1 \to 2} = V(2) - V(1) \approx 0.658$. Conversely, the energy required to escape the true vacuum (U=3) is drastically higher, $\Delta V_{3 \to 2} = V(2) - V(3) \approx 1.804$. Any perturbation analysis on the continuous PDE interpretation must account for this extreme asymmetry, which aggressively favors the total invasion of the domain by the U=3 state.

4. Continuous PDE Dynamics: Front Propagation and Heteroclinic Connections

Transitioning from the local ODE kinetics and scalar potential to the full continuous PDE interpretation, the introduction of spatial diffusion (D=0.12) breaks the isolated local symmetry and allows for continuous traveling wave solutions. In a one-dimensional infinite continuous domain, the system admits traveling front solutions of the form $U(x,t) = \Psi(z)$, where the co-moving coordinate is $z = x - ct$, and c is the asymptotic wave speed.
Substituting this traveling wave ansatz into the unperturbed continuous PDE yields a second-order nonlinear ordinary differential equation acting as an eigenvalue problem for the wave speed c:
In applied mathematics, this equation elegantly describes a classical mechanics analogy: a Newtonian particle with mass D=0.12 moving in the reversed potential $-V(\Psi)$ with spatial friction proportional to the unknown wave speed c. The continuous PDE interpretation requires finding a specific spatial friction coefficient c that permits a heteroclinic orbit in the (Ψ, Ψ') phase plane. This orbit must perfectly connect the saddle points of the reversed potential, which mathematically correspond to the stable fixed points (U=1 and U=3) of the local ODE kinetics. Because $V(3) < V(1)$, a unique, stable traveling front exists connecting U=3 at $z \to -\infty$ to U=1 at $z \to +\infty$. Multiplying the traveling wave equation by the spatial gradient Ψ' and integrating across the entire infinite domain yields the exact energetic balance for the wave speed c:
The right-hand side of this equation represents the total thermodynamic driving force. Based on our exact potential derivations, $V(1) - V(3) \approx (-0.1142) - (-1.2601) = 1.1459$. Since the driving force is strictly positive, and the integral of the squared gradient $\int (\Psi')^2 dz$ is inherently strictly positive, the spatial wave speed c must be strictly positive.
In the continuous PDE interpretation, this constitutes a mathematical proof that the state U=3 will continuously, monotonically, and inexorably invade the state U=1. The state U=1 cannot dynamically invade U=3 under any initial conditions in the unperturbed continuous PDE limit; any initial spatial domain containing U=3 that exceeds a specific critical spatial threshold will trigger a self-sustaining propagating front that totally subsumes the U=1 phase. This transition from a bistable ignition regime to a monostable propagation regime is a hallmark of asymmetric gradient-driven reaction-diffusion systems.

5. Multi-Dimensional PDE: Classical Nucleation Theory and the Curvature Penalty

When the continuous PDE interpretation is extended from a one-dimensional line to two or three spatial dimensions ($d \ge 2$), the dynamics of front propagation are heavily modified by geometric constraints. The flat heteroclinic front defined in 1D is penalized by mean curvature,

bringing the system under the rigorous governance of Classical Nucleation Theory (CNT).
If a localized inclusion (a "droplet") of the true vacuum ($U=3$) is placed inside a vast domain of the false vacuum ($U=1$), the interface experiences a severe inward radial force due to surface tension. This surface tension is mathematically driven by the gradient penalty term $\frac{D}{2}|\nabla U|^2$ in the free-energy functional. The continuous PDE interpretation dictates that the expansion or collapse of the $U=3$ droplet depends entirely on the competition between the volumetric energy gain (the thermodynamic driving force $\Delta V_{drive} = V(1) - V(3)$) and the interfacial energy cost.
The total excess free energy of a d-dimensional droplet of radius R is approximately:
where α_d and β_d are geometric volume and surface area coefficients, and σ is the interface surface tension defined by the integral across the heteroclinic orbit: $\sigma = \int_{1}^{3} \sqrt{2D(V(U)-V_{min})}\, dU$.
The nucleation barrier is mathematically formalized as the critical radius R_c, found by maximizing the free energy $\frac{\partial \Delta \mathcal{E}}{\partial R} = 0$. If the initial perturbation has a radius $R < R_c$, the diffusion operator dominates the polynomial reaction, surface tension violently collapses the droplet, and the local ODE kinetics drive the system back to the uniform $U=1$ false vacuum. If $R > R_c$, the volumetric energy advantage overcomes the mean curvature penalty, allowing the $U=3$ state to autocatalytically nucleate and propagate outward asymptotically approaching the 1D wave speed c.
The exact critical radius in the continuous PDE interpretation can be approximated via the thin-wall limit (valid when the interface thickness is much smaller than the droplet radius):
Consequently, while the local ODE kinetics view $U=2$ merely as a simple, zero-dimensional unstable point, the continuous PDE interpretation recognizes a massive multidimensional nucleation barrier that demands a spatially extended, highly specific energetic perturbation to cross the threshold into the $U=3$ basin. The pre-exponential kinetic factors (e.g., the Zeldovich factor) further dictate that thermally or numerically induced random fluctuations are overwhelmingly unlikely to spontaneously bridge this gap without directed initial conditions.

6. Discretized Finite-Grid Implementation: Lattice Constraints and Propagation Failure

The most severe divergence between the theoretical applied mathematics analysis and practical computational simulation occurs when evaluating spatial discretization. It is critical to explicitly state that the phenomenon of continuous, smooth wave propagation, proven in Section 4 for the continuous PDE interpretation, frequently fails catastrophically in the discretized finite-grid implemented system due to the breakdown of continuous translational invariance.
When the continuous spatial operator $D\nabla^2 U$ is replaced by a discrete numerical Laplacian (for example, the standard second-order central difference $\frac{D}{\Delta x^2} (U_{i+1} - 2U_i + U_{i-1})$), the PDE is fundamentally transformed into a Lattice Differential Equation (LDE). In this discretized finite-grid implementation, the moving front must overcome a series of periodic energetic hurdles known as the Peierls-Nabarro barrier, which is an artifact of the mesh geometry itself.
Because the diffusion coefficient is locked at a specifically low value ($D=0.12$), the continuous spatial coupling is inherently weak. This weak diffusion naturally compresses the interface thickness of the traveling front. If the spatial discretization step Δx is not infinitesimally small, the coupling ratio $\frac{D}{\Delta x^2}$ drops below a critical mathematical threshold. When this occurs, the thermodynamic driving force of the effective potential $V(1) - V(3)$ becomes insufficient to pull the next discrete grid node over the local ODE nucleation barrier at $U=2$.
The result is a phenomenon known as **propagation failure** or **front pinning**. In this depinning transition regime, the traveling front completely halts, becoming a static, stepped pattern frozen

across the grid. This is entirely an artifact of the discretized finite-grid implemented system; the continuous PDE interpretation explicitly denies the existence of stationary fronts when an energetic imbalance $V(1) \neq V(3)$ exists.

An unaware observer running a simulation with $\Delta x \geq 0.5$ might falsely conclude that the local ODE kinetics possess an artificial, heightened stability around $U=1$, or that the continuous PDE lacks the requisite energy to propagate. In reality, the discretized finite grid has trapped the heteroclinic orbit in a structurally induced periodic local minimum, creating an empirical observation of stability that is mathematically fictitious in the continuum limit.

7. The Renewal Term $\Gamma(U)$: Perturbations and the Destruction of Gradient Topology

The inclusion of the undefined renewal operator $\Gamma(U)$ presents a substantial vulnerability to the topological integrity of the system established thus far. Without an explicit definition of $\Gamma(U)$, its behavior must be bounded through rigorous perturbation theory applied sequentially to both the local ODE kinetics and the continuous PDE interpretation.

If $\Gamma(U)$ is defined as a strictly autonomous, localized, and zero-dimensional perturbation (for example, a simple linear decay $\Gamma(U) = -\epsilon U$), it immediately breaks the exact integer symmetries of the $\sin(\pi U)$ roots. The mathematical fixed points of the local ODE kinetics would shift off the exact integer values $U=1, 2, 3$, generating transcendental roots. A sufficiently large magnitude for $\Gamma(U)$ could completely annihilate the false vacuum near $U=1$ via a saddle-node bifurcation, merging it destructively with the unstable $U=2$ manifold. This would irrevocably destroy the bistability, reducing the continuous PDE interpretation from a rich multi-basin landscape into a monostable ignition equation.

Alternatively, and far more dangerously, if $\Gamma(U)$ is an integro-differential non-local term (such as spatial averaging $\int K(x-y)U(y)dy$ to simulate resource depletion) or incorporates an explicit time-delay $\Gamma(U(t-\tau))$, it fundamentally shatters the gradient flow structure established in Section 3. A time-delayed $\Gamma(U)$ categorically invalidates the existence of the scalar effective potential $V(U)$.

While the unperturbed system exhibits a negative maximum Lyapunov exponent guaranteeing strict convergence to a steady state , a delayed or spatially non-local renewal term could force the continuous PDE into entirely new dynamical regimes. These include Hopf bifurcations, breathing fronts, chaotic spatiotemporal intermittency, or oscillatory Turing-like spatial patterns. However, strictly enforcing the rigorous parameters provided in the prompt—and excluding artificial assumptions of sustained chaos—requires treating $\Gamma(U)$ strictly as an asymptotically decaying or highly bounded perturbation that preserves the underlying gradient topology, rather than a mechanism for exotic chaotic generation.

8. The Non-Negativity Clamp: Implementation Artifacts and Conservation Violations

The explicit requirement to enforce a "numerical clamp" ($U \geq 0$) is a purely implementation-level empirical mechanism. It is fundamentally absent from both the exact local ODE kinetics and the rigorous continuous PDE interpretation, and its presence indicates a concession to numerical instability.

In a pristine continuous PDE, if the system approaches $U=0$ from the right, the reaction term evaluates to $f(U) \approx 0.45\pi U^3$, which is strictly positive. The continuum mathematics continuously drive the system away from negative values without requiring external intervention. However, discretized numerical integration schemes (such as explicit forward Euler or low-order

Runge-Kutta methods) operating with finite, discrete time steps (Δt) inevitably suffer from truncation error and overshoot. When numerical diffusion pushes a node across the axis to $U < 0$, the term $\sin(\pi U)$ violently flips sign, creating an artificial, unphysical oscillation that rapidly and catastrophically destabilizes the solver.
Applying a hard clamp ($U_{i}^{n+1} = \max(0, U_{i}^{n+1})$) is a computationally crude, mathematically non-smooth intervention. In the discretized finite-grid implemented system, this clamp acts as a continuous, unbounded injector or extractor of mass, violating the underlying conservation laws of the PDE. Whenever the clamp triggers, the divergence theorem mapping fluxes across the control volumes is mathematically broken.
Furthermore, the clamp forcefully introduces an artificial absorbing boundary in the phase space. While the local ODE kinetics correctly identify $U=0$ as a non-hyperbolic repulsive fixed point with a delicate algebraic escape trajectory, the implementation-level clamp artificially pins numerical noise to exactly zero. This fundamentally distorts the low-density asymptotic tails of any traveling front and alters the leading-edge physics of the wave propagation. A rigorous numerical implementation should instead utilize structure-preserving, unconditionally positive implicit solvers rather than a brute-force clamp that corrupts the analytical fidelity of the continuous PDE.

9. Spectral Stability and the Strict Exclusion of Sustained Chaos

Addressing the strict injunction to exclude hype regarding sustained chaos requires formally examining the system's spectral stability and maximal Lyapunov exponents. In dynamical systems theory, sustained chaos requires continuous stretching and folding within the phase space, a phenomenon mathematically indicated by a strictly positive maximum Lyapunov exponent ($\lambda_{max} > 0$).
Because the core reaction-diffusion mechanism (excluding pathological, non-gradient formulations of $\Gamma(U)$) relies entirely on a single scalar field driven by an exact effective potential $V(U)$, the continuous PDE interpretation represents a strict gradient dynamical system. Gradient systems are topographically and mathematically prohibited from exhibiting chaotic attractors, strange attractors, or even stable limit cycles (homoclinic tangles notwithstanding). The trajectory must always flow downhill in the potential landscape.
Calculations of the Jacobian eigenvalues across the entire spatial domain of this system will consistently reveal a spectrum tightly bounded in the left half of the complex plane, yielding a strictly negative maximum Lyapunov exponent ($\lambda_{max} < 0$). Any transient irregularity, complex pattern formation, or pseudo-chaotic behavior observed during numerical simulations will ultimately collapse into a stable, stationary spatial pattern, a pinned front, or a uniform homogeneous state.
If chaotic fluctuations are observed in the discretized finite-grid implemented system, they are mathematically guaranteed to be the result of one of three artifacts: 1) numerical instability from exceeding the Courant–Friedrichs–Lewy (CFL) limit, 2) artificial truncation noise injected by the non-negativity clamp, or 3) spurious high-frequency eigenvalues injected by inappropriate spatial mesh configurations. Intrinsic, mathematically persistent chaos is an impossibility within the continuum topology of this specific locked-parameter system.

10. Conclusion and Vulnerability Matrix

This audit has rigorously deconstructed the specific applied mathematics formulation $\partial U/\partial t = 0.12 \nabla^2 U + 0.45 U^2 \sin(\pi U) + \Gamma(U)$. The fundamental structure of

the system hinges entirely on recognizing the distinct mathematical behaviors across three distinct modeling domains. The integer levels U=1, U=2, and U=3 are strictly proven to be exact mathematical fixed points of the unperturbed local ODE kinetics, dictating a thermodynamic topology of false and true vacua.

The continuous PDE interpretation relies upon an analytically verifiable exact effective potential, confirming that U=3 possesses total global dominance and will relentlessly invade the U=1 state. However, the practical realization of this system is fiercely contested by the discretized finite-grid implemented system, where weak diffusion (D=0.12) threatens macroscopic front pinning, and the implementation-level non-negativity clamp destroys exact mass conservation. Any study of this system that fails to rigorously isolate these three domains will inevitably misattribute numerical artifacts to physical phenomena, leading to fundamentally compromised conclusions regarding nucleation thresholds and wave propagation.

Summary Lines:

1. The states U \approx 1, 2, 3 are exact mathematical fixed points of the local ODE kinetics, dictating an asymmetrical true/false vacuum topology governed by an explicitly derivable effective potential gradient.
2. In the continuous PDE interpretation, this gradient guarantees the persistent invasion of the U=1 state by the U=3 state via heteroclinic front propagation, strictly precluding chaotic attractors due to a negative maximum Lyapunov exponent.
3. However, in the discretized finite-grid implemented system, the weak D=0.12 parameter risks severe macroscopic wave pinning, while the implementation-level U \geq 0 clamp destroys exact mass conservation.

Final Methodological Mapping Table:

Claim I might want to make	Narrowest version I can actually defend	Why the stronger version would be vulnerable
The system naturally evolves to a sustained, chaotic spatial pattern.	The transient dynamics exhibit pattern formation that strictly decays to steady-state attractors with a negative maximum Lyapunov exponent.	The continuous PDE is mathematically proven to be a gradient flow (via the effective potential V(U)), categorically preventing sustained chaotic attractors.
The stable states U=1 and U=3 form a perfectly symmetric bistable landscape.	The local ODE yields stable fixed points at exactly U=1 and U=3, but the effective potential reveals U=3 as the dominant global minimum.	Assuming symmetric bistability ignores the thermodynamic energetic imbalance \Delta V \approx 1.146, fundamentally mischaracterizing the inevitable direction of continuous PDE front invasion.
The continuous front will always reliably propagate and engulf the domain.	The continuous PDE interpretation guarantees front propagation, provided the initial droplet exceeds the multidimensional nucleation barrier R_c.	The discretized finite-grid implementation introduces Peierls-Nabarro lattice barriers, which can physically freeze the front (propagation failure) due to the low D=0.12 coupling.
The non-negativity clamp (U \ge 0) ensures physical realism without altering the physics.	The non-negativity clamp is a necessary implementation-level heuristic to prevent explicit solver instability caused by numerical overshoot near the non-hyperbolic U=0 fixed point.	The clamp breaks the continuous PDE conservation laws, injecting artificial mass and transforming a smooth continuous topology into a discontinuous, absorbing

Claim I might want to make	Narrowest version I can actually defend	Why the stronger version would be vulnerable
		numerical boundary.

Works cited

1. DEPINNING TRANSITIONS IN DISCRETE REACTION-DIFFUSION EQUATIONS - Docta Complutense, https://docta.ucm.es/bitstreams/26b811a9-a7f1-4c24-b847-7ed494934c5d/download 2. Sharp transition between extinction and propagation of reaction - UCSD Math, https://mathweb.ucsd.edu/~zlatos/Papers/sharpquen.pdf 3. Boosting of neural circuit chaos at the onset of collective oscillations - eLife, https://elifesciences.org/reviewed-preprints/90378 4. arXiv:0902.2545v1 [nlin.CD] 15 Feb 2009, https://arxiv.org/pdf/0902.2545 5. Introduction to Mathematical Modeling and Chaotic Dynamics [1st ed.] 9781439898864 - DOKUMEN.PUB, https://dokumen.pub/introduction-to-mathematical-modeling-and-chaotic-dynamics-1stnbsped-9781439898864.html 6. (PDF) From Newton to Cellular Automata - ResearchGate, https://www.researchgate.net/publication/45889387_From_Newton_to_Cellular_Automata 7. The mechanics of nucleation and growth and the surface tensions of active matter - OSTI, https://www.osti.gov/servlets/purl/2587604 8. Exact travelling wave solutions of an “integrable” discrete reaction, https://spiral.imperial.ac.uk/entities/publication/109383d6-1ebc-4eb7-9ee7-5808d1122106 9. Control of clustered action potential firing in a mathematical model of entorhinal cortex stellate cells - PMC, https://pmc.ncbi.nlm.nih.gov/articles/PMC5947116/ 10. Stochastic transitions in a bistable reaction system on the membrane - PMC - NIH, https://pmc.ncbi.nlm.nih.gov/articles/PMC3673150/ 11. Existence and non-existence of transition fronts for bistable and ignition reactions - Numdam, https://www.numdam.org/item/AIHPC_2017__34_7_1687_0.pdf 12. Quantitative Estimates of the Threshold Phenomena for Propagation in Reaction-Diffusion Equations | SIAM Journal on Applied Dynamical Systems, https://epubs.siam.org/doi/10.1137/19M1292187 13. A Review of Classical and Nonclassical Nucleation Theories | Crystal Growth & Design, https://pubs.acs.org/doi/10.1021/acs.cgd.6b00794 14. Classical nucleation theory - Wikipedia, https://en.wikipedia.org/wiki/Classical_nucleation_theory 15. State-dependent diffusion coefficients and free energies for nucleation processes from Bayesian trajectory analysis - PMC, https://pmc.ncbi.nlm.nih.gov/articles/PMC6171659/ 16. Nucleation barrier reconstruction via the seeding method in a lattice model with competing nucleation pathways | The Journal of Chemical Physics | AIP Publishing, https://pubs.aip.org/aip/jcp/article/145/21/211912/196117/Nucleation-barrier-reconstruction-via-the-seeding 17. Cortical hot spots and labyrinths: why cortical neuromodulation for episodic migraine with aura should be personalized - Frontiers, https://www.frontiersin.org/journals/computational-neuroscience/articles/10.3389/fncom.2015.00029/full 18. Nucleation - PMC - NIH, https://pmc.ncbi.nlm.nih.gov/articles/PMC2995260/ 19. Numerical studies of nucleation pathways of ordered and disordered phases - UvA-DARE (Digital Academic Repository), https://pure.uva.nl/ws/files/4183980/52806_valeriani.pdf 20. Tropical tree cover in a heterogeneous environment: A reaction-diffusion model | PLOS One, https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0218151 21. (PDF) Traveling and pinned fronts in bistable reaction-diffusion systems on network, https://www.researchgate.net/publication/386825945_Traveling_and_pinned_fronts_in_bistable_reaction-diffusion_systems_on_network 22. [PDF] Depinning Transitions in Discrete Reaction-Diffusion Equations | Semantic Scholar, https://www.semanticscholar.org/paper/Depinning-Transitions-in-Discrete-Equations-Carpio-Bon

illa/67c91ff0f28c113a2730f27d4cdb11ec8a433257 23. Propagation Failure in Discrete Inhomogeneous Medium Using a Caricature of the Cubic - ucf stars, https://stars.library.ucf.edu/cgi/viewcontent.cgi?article=2227&context=etd 24. Variational approximation and the use of collective coordinates, https://people.bath.ac.uk/jhpd20/publications/DawesSusanto-PRE2013.pdf 25. From actin waves to mechanism and back: How theory aids biological understanding - PMC, https://pmc.ncbi.nlm.nih.gov/articles/PMC10332813/ 26. Propagation failure in discrete bistable reaction-diffusion systems: Theory and experiments, https://www.researchgate.net/publication/11846585_Propagation_failure_in_discrete_bistable_reaction-diffusion_systems_Theory_and_experiments 27. Multi-front dynamics in spatially inhomogeneous Allen-Cahn equations, https://pub.math.leidenuniv.nl/~doelmana/BDK25-submitted.pdf 28. Reaction–diffusion equations in the half-space - EMS Press, https://ems.press/content/serial-article-files/25255 29. Propagation or extinction in bistable equations: The non-monotone role of initial fragmentation - American Institute of Mathematical Sciences, https://www.aimsciences.org/article/doi/10.3934/dcdss.2023165 30. (PDF) Blow-up for a non-local diffusion problem with Neumann boundary conditions and a reaction term - ResearchGate, https://www.researchgate.net/publication/222813897_Blow-up_for_a_non-local_diffusion_problem_with_Neumann_boundary_conditions_and_a_reaction_term 31. A local Echo State Property through the largest Lyapunov exponent - ResearchGate, https://www.researchgate.net/publication/260107552_A_local_Echo_State_Property_through_the_largest_Lyapunov_exponent 32. Optimal Sequence Memory in Driven Random Networks - Fraunhofer-Publica, https://publica.fraunhofer.de/bitstreams/88e6a447-a707-43e7-81ba-b864600f9de4/download 33. Wave Propagation Phenomena in Nonlinear Hierarchical Neural Networks with Predictive Coding Feedback Dynamics - PMC, https://pmc.ncbi.nlm.nih.gov/articles/PMC12289805/ 34. arXiv:2007.02767v1 [nlin.PS] 6 Jul 2020, https://arxiv.org/pdf/2007.02767 35. Innovative Solutions for Data Accessibility: Utilizing Synthetic Data in DEWA Operations - RIT Digital Institutional Repository, https://repository.rit.edu/cgi/viewcontent.cgi?article=13303&context=theses 36. A systematic study of autonomous and nonautonomous predator–prey models for the combined effects of fear, refuge, cooperation and harvesting - PMC, https://pmc.ncbi.nlm.nih.gov/articles/PMC9217126/ 37. Cellular dynamics and stable chaos in balanced networks - eDiss, https://ediss.uni-goettingen.de/bitstream/handle/11858/00-1735-0000-0028-869D-B/thesis2publish.pdf?sequence=1&isAllowed=y 38. Flow Matching for Few-Trial Neural Adaptation with Stable Latent Dynamics | OpenReview, https://openreview.net/forum?id=nKJEAQ6JCY